

NAM TRUNG TRAN - NATHAN TRAN

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EDUCATION

Adelaide University, Australia (Global Citizens Scholarship)

2023 - 2026

Bachelor of Computer Science, Major in Artificial Intelligence

GPA: 6.7/7.00

EXPERIENCE

AI Engineer Intern

Dec 2025 - Feb 2026

Viettel Solutions Corporation

Hanoi, Vietnam

- Fine-tuned and benchmarked natural language processing models, measuring performance across multiple task types to compare candidate approaches
- Designed and ran testing and evaluation procedures for an AI product line, verifying output reliability and accuracy before release
- Worked with senior engineers to analyse results and translate findings into recommendations for scalable AI deployment

Summer Research Assistant

Dec 2024 - Feb 2025

SET Faculty, University of Adelaide

Adelaide, SA, Australia

- Collaborated with Professor Hung Nguyen on automating MITRE ATT&CK flow graph generation from Cyber Threat Intelligence reports, leveraging Large Language Models to improve efficiency and accuracy
- Validated research reliability by rerunning 20+ experimental outputs, ensuring reproducibility of findings and conducted comparative evaluation of Gemini 1.5 Pro and Gemini 1.5 Flash models

Datathon – 1st place

April 2025

SQUAD x insightfactory.ai/18-hour competition

Adelaide, SA, Australia

- Collaborated in a 3-person team to tackle an anonymized dataset with 19 features and an unlabelled target variable, conducting exploratory data analysis, feature engineering and visualization to uncover insights and patterns
- Developed and evaluated machine learning models under time constraints, applying data cleaning, preprocessing and validation techniques to deliver top-performing solution

PROJECTS

NBA Fair-Value Model | [live demo](#) | *Python, scikit-learn, Streamlit, pytest*

July 2026

- Built a gradient-boosted model predicting an NBA player's fair salary from on-court performance alone, trained on six seasons and roughly 2,100 player-seasons scraped from Basketball-Reference and ESPN
- Achieved R^2 of 0.73 under leave-one-season-out cross-validation, holding between 0.62 and 0.78 across every held-out season
- Diagnosed and removed target leakage in the original model, adding a unit test that fails the build if any salary-derived column re-enters the feature matrix; deployed an interactive Streamlit explorer with player lookup and over/underpaid leaderboards

Auto-Updating VISA Intelligence System | [live demo](#) | *Python, ChromaDB, Llama 3.3*

Apr. 2026 – Present

- Built a production RAG pipeline over live Australian immigration content, rendering JavaScript pages in headless Chromium via Playwright, filtering and chunking them into 768-dimension embeddings in a ChromaDB index, and answering questions with Llama 3.3 70B constrained to retrieved sources
- Lifted top-1 retrieval accuracy from 50% to 80%, and top-3 from 70% to 100%, across a 10-question benchmark by diagnosing a vocabulary mismatch between applicant phrasing and government wording, then adding LLM query expansion with reciprocal rank fusion
- Raised top-1 accuracy a further 20 points by profiling the corpus and finding generic application forms occupied 246 of 478 indexed chunks while never mentioning the visa; filtered them on content rather than filename
- Hardened the daily ingestion job to 26 consecutive successful runs scraping 4/4 sources, and expanded the test suite from 100 to 213 tests including integration coverage against a real vector index

Study1Only | *Vue.js, Node.js, Express, MySQL, Spotify API*

Feb. 2025 – June 2025

- Designed the relational database schema and implemented the achievement and streak system end to end for a deployed study-tracking platform built by a team of four
- Built the admin dashboard for user administration, and served as the team's tester across feature integration

SKILLS

Languages: Python, C/C++, SQL, JavaScript/TypeScript, HTML/CSS, R, MATLAB

Machine Learning: scikit-learn, pandas, NumPy, PyTorch, LangChain, ChromaDB, RAG

Web & Databases: Node.js, Express, Vue.js, Next.js, REST APIs, PostgreSQL, MySQL, SQLite

Tools: Git, GitHub Actions, pytest, Streamlit, Docker, Jupyter